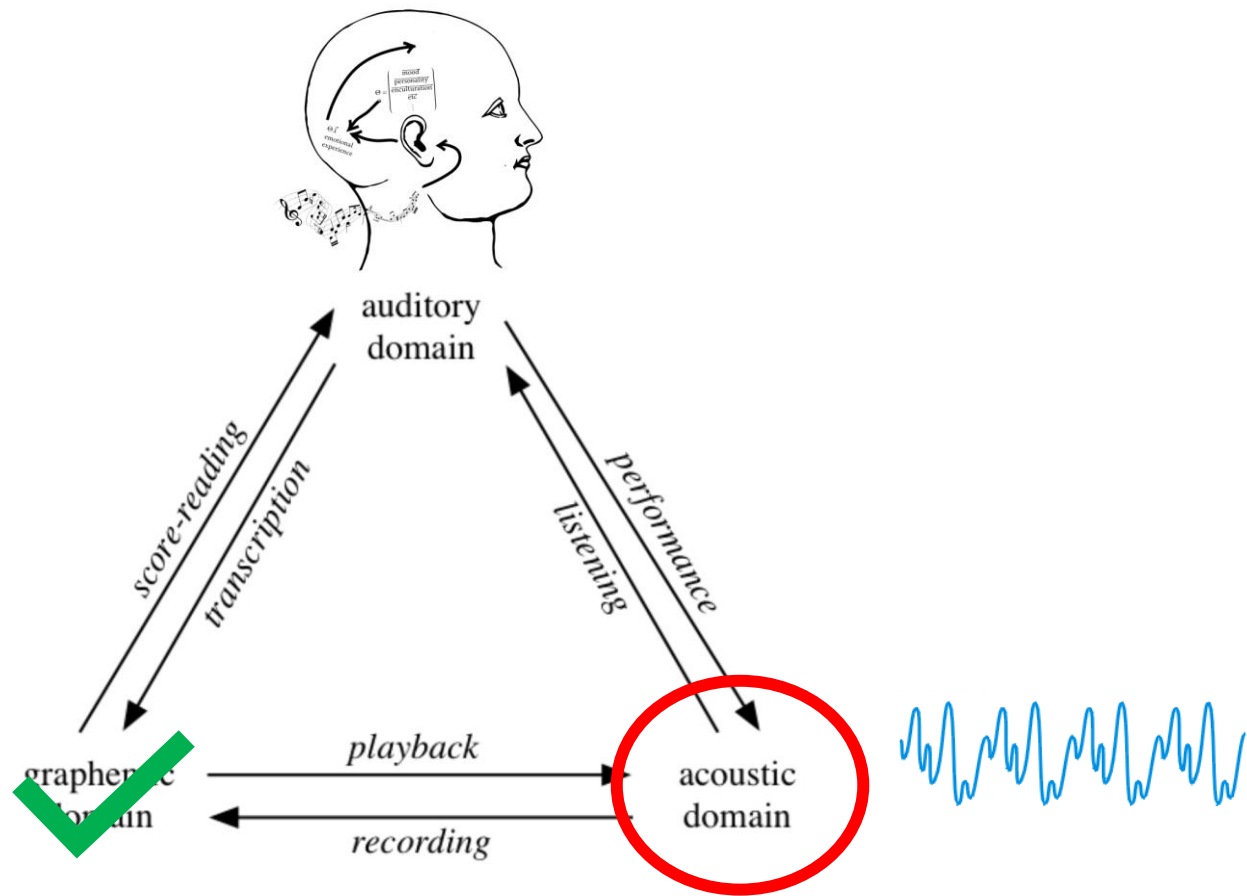


# Audio Processing in Digital Musicology

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On the non-existence of music: Why music theory is a figment of the imagination. G. A. Wiggins et al. (2010), *Musicae Scientiae, Discussion Forum*, 5, pp. 231-255 \

Babbitt, M. (1965). The use of computers in musicological research. *Perspectives of New Music*, 3(2), 74-83.

**When audio?**

## Arguments for audio

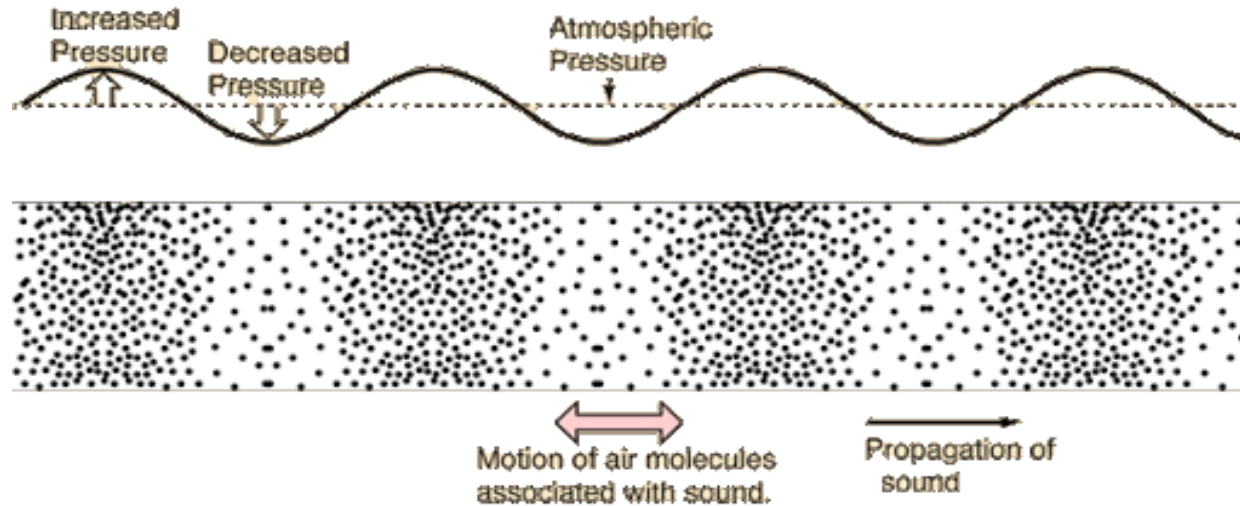
- Arguably, the original source representation in which music exists
- There is no notation, or it is not very informative, for many musical genres (electronic, ethnikal music, pop, improvisations)
- Allows to study timbre and production style
- Audio is what the listeners actually hear (emotional response, perception, popularity)

## When not to use audio

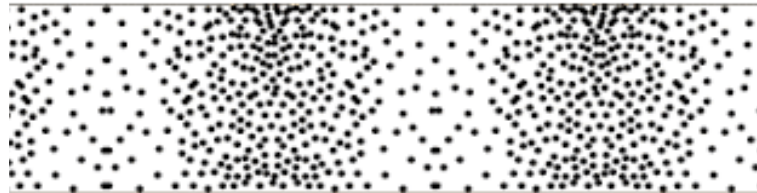
- Your question is best answerable with a score (usually, involves studying the composition process)
- You need precision that is inachievable with state-of-the-art audio processing (polyphonic transcription of orchestra pieces)
- The audio recordings you have are low-quality (old technology, live recordings)
- Audio processing can introduce artifacts, so interpretive caution is necessary

# **Audio representations and feature extraction**

*Sound is an oscillation of pressure transmitted through gases, plasma, and liquids as longitudinal waves and through solids as both longitudinal waves and transverse waves.*



# How can we write it down?



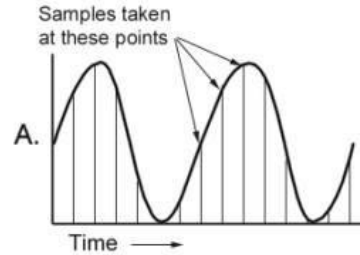
↔  
Oscillation of air molecules  
associated with sound.

→  
Propagation of sound

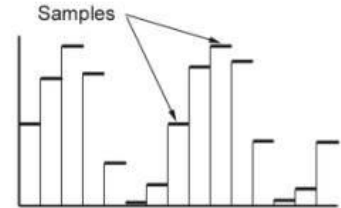


## Increasing Sample Rates

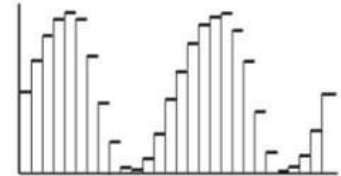
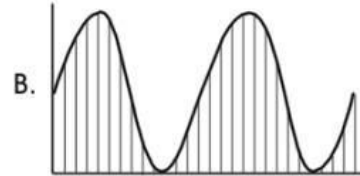
Analog Wave



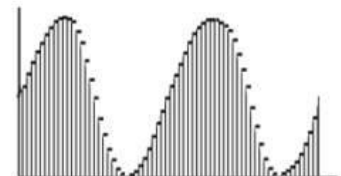
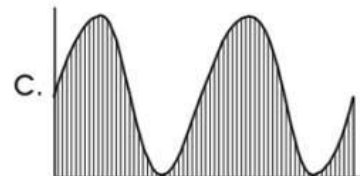
Digital Result



=



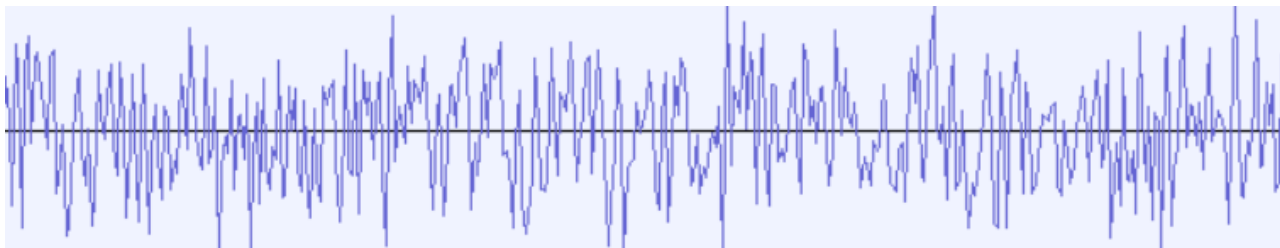
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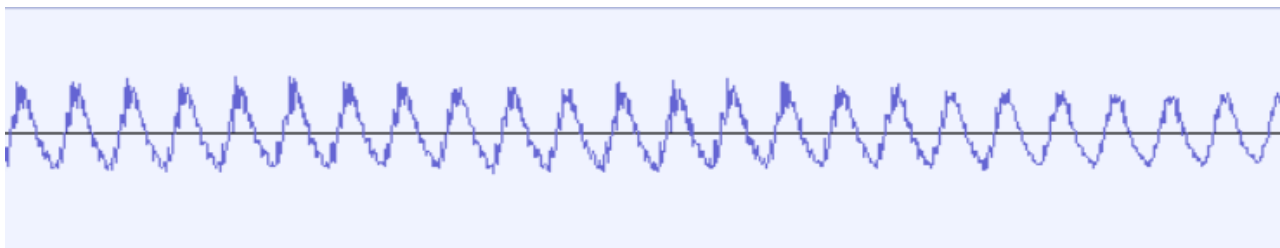
=



(1)



(2)



*Why do some sounds look like (1) and others like (2)?*

## Tone vs noise

### TONAL SOUNDS

A signal that repeats systematically and continuously over time

Examples:

- Mammalian vocalizations
- Bird song
- Musical instruments
- Sine wave

### NOISE

Sounds over all the frequency range can be heard at random with equal probability

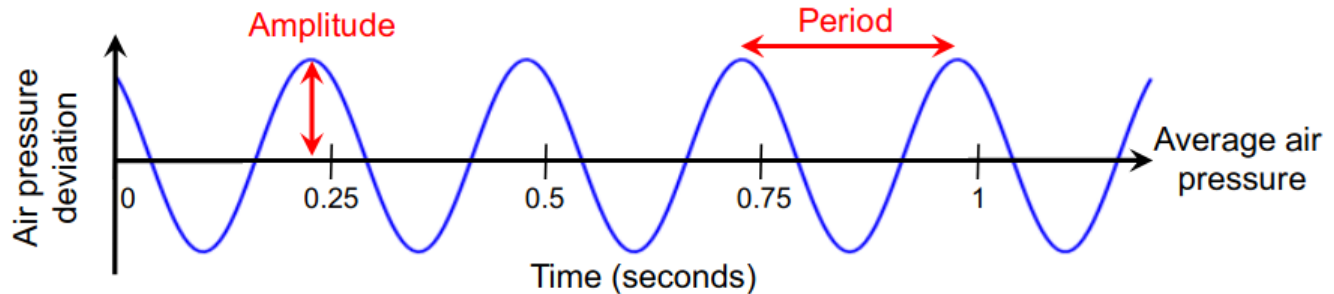
White noise is an extreme example

Many natural sound sources are noise-like:

- Waterfall
- Rain
- Rustling of the leaves

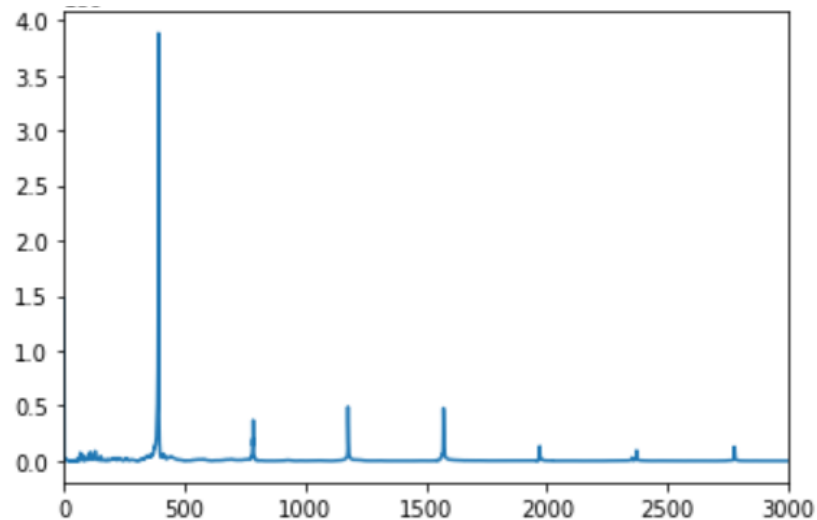
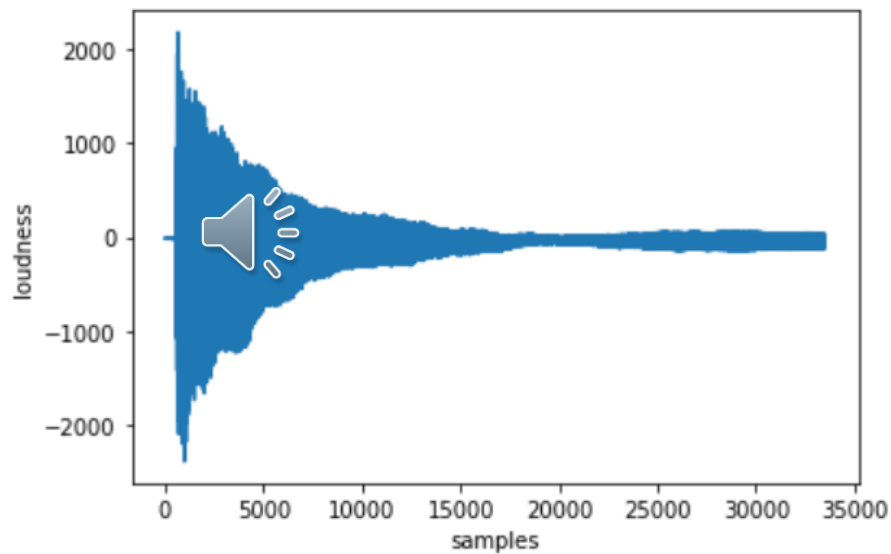
# Basic properties of sound

- Amplitude (loudness)
- Frequency (pitch)
- Spectrum and dynamical envelope (timbre)
- Duration (length)

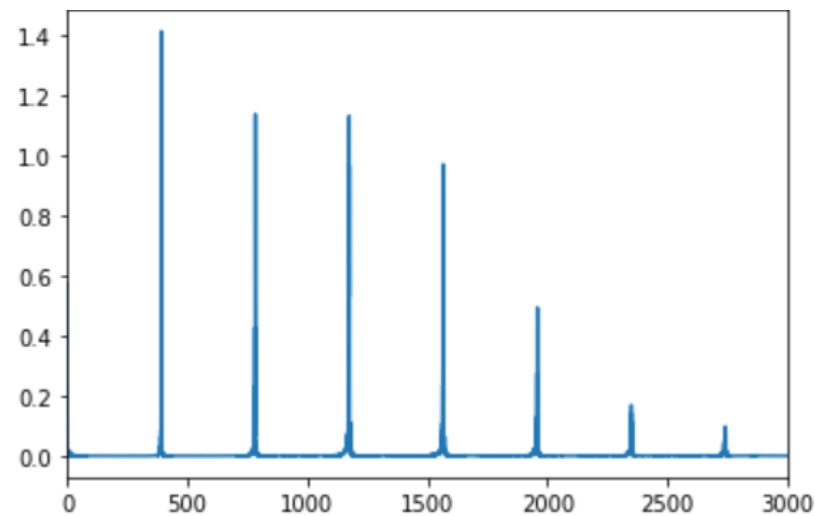
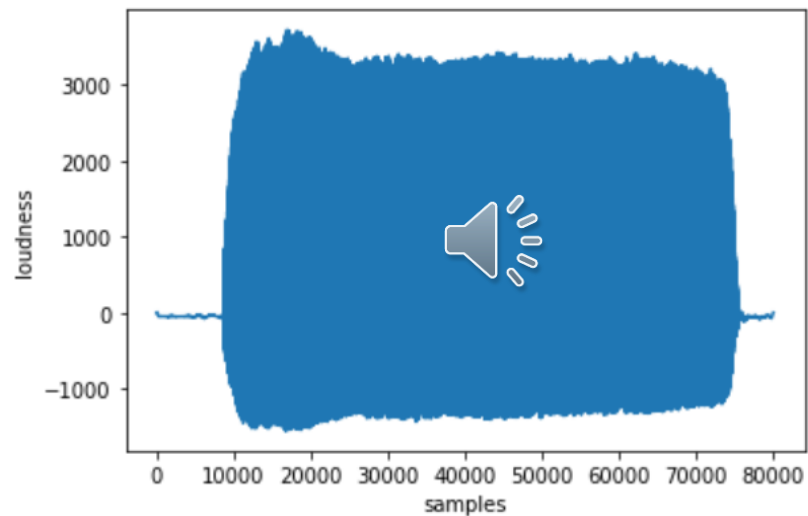


# Translation to music

## Piano - G4 note



# Trumpet - G4 note



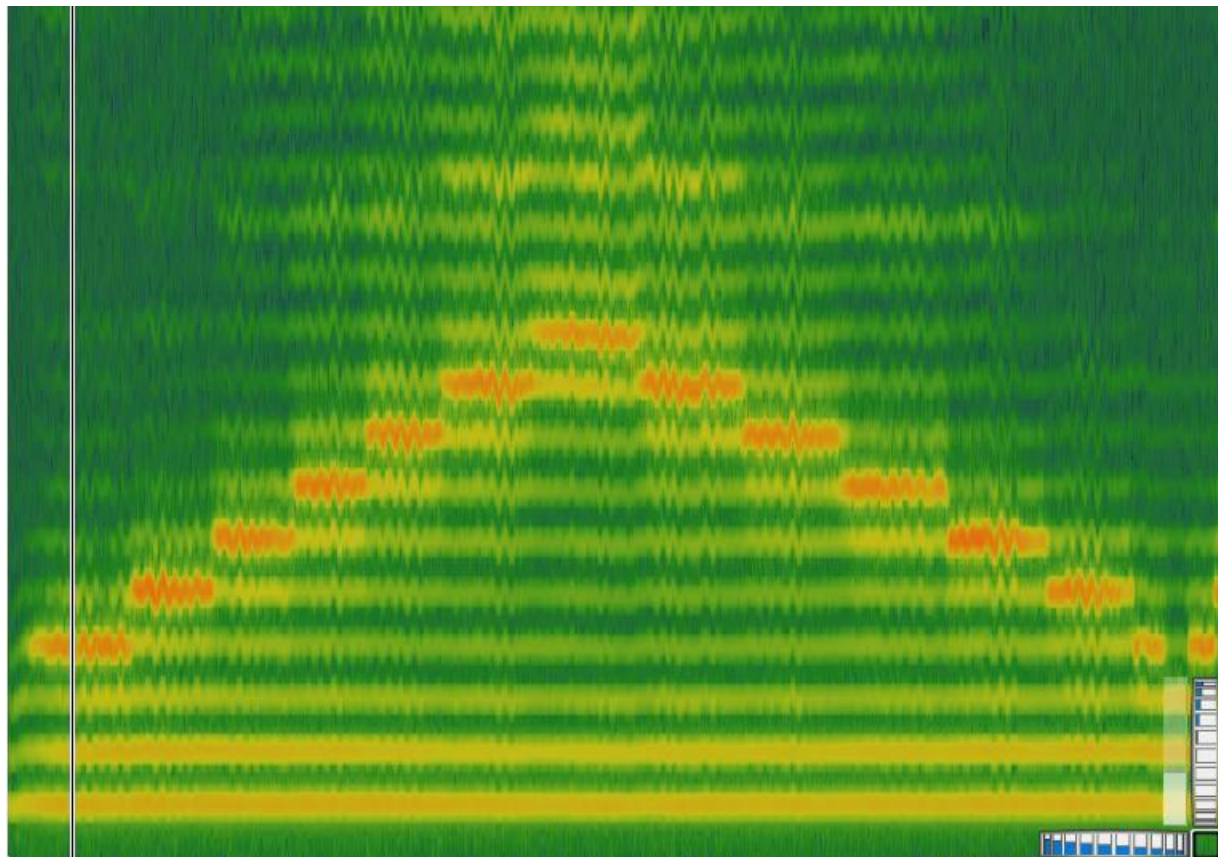
# Timbre

- Timbre is the color of the sound
- Timbre is defined by:
  - Sound envelope
  - Frequency spectrum
  - Fluctuations of the frequency spectrum



Anna-Maria Hefe: Overtone Singing

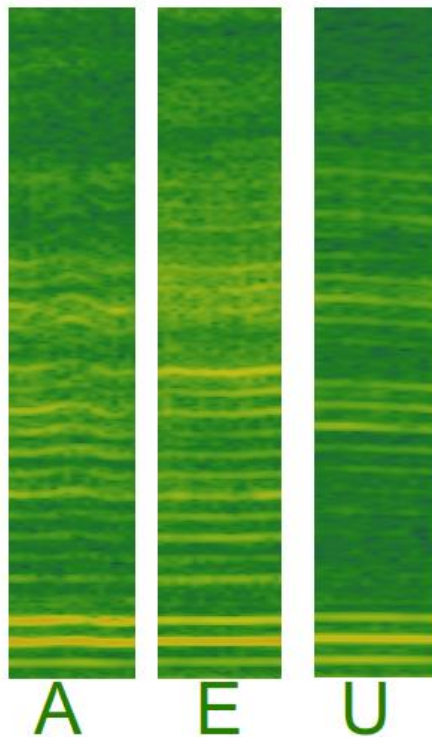




*The same thing: spectrogram representation*

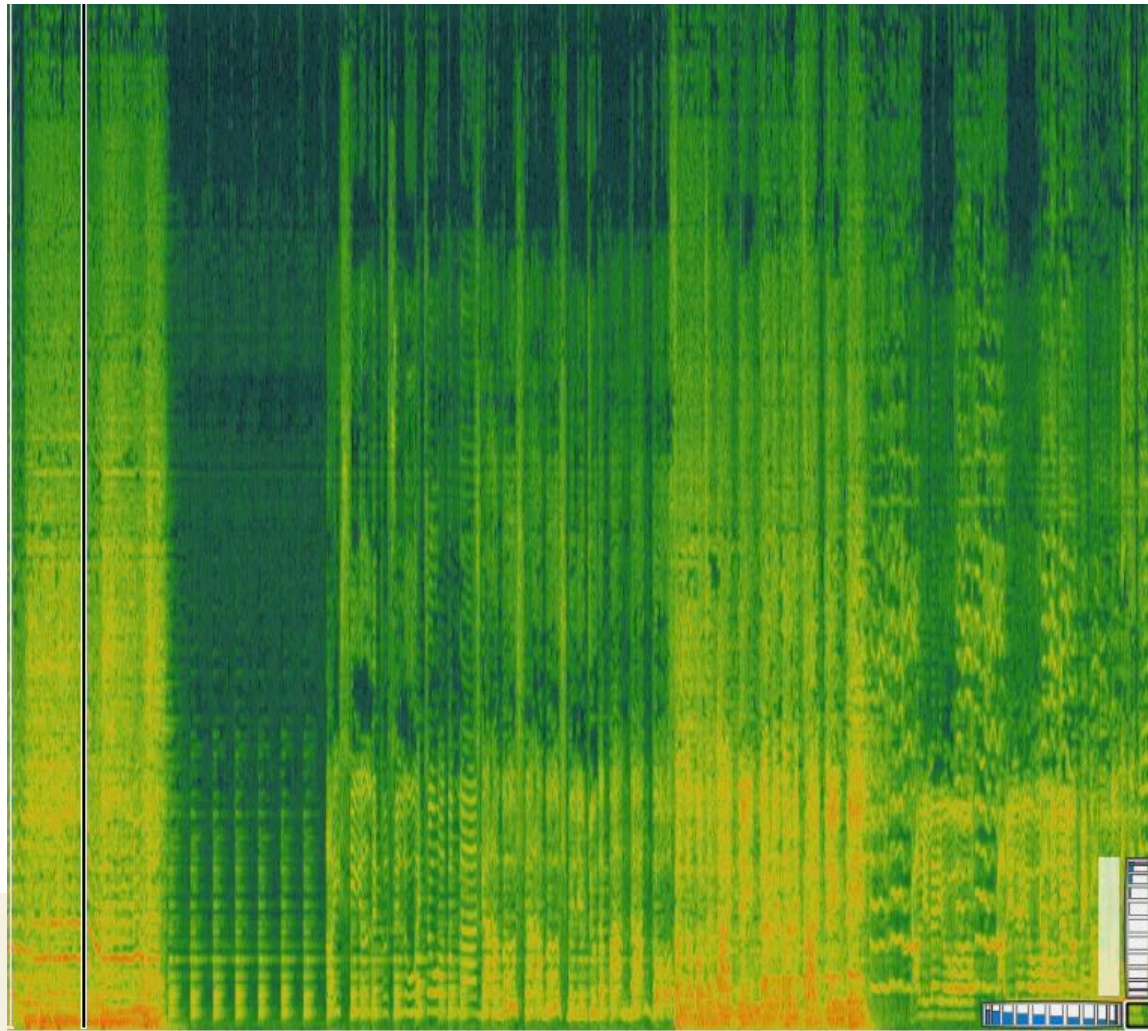
Can you magnify certain  
overtones in your voice?

Everyone can!



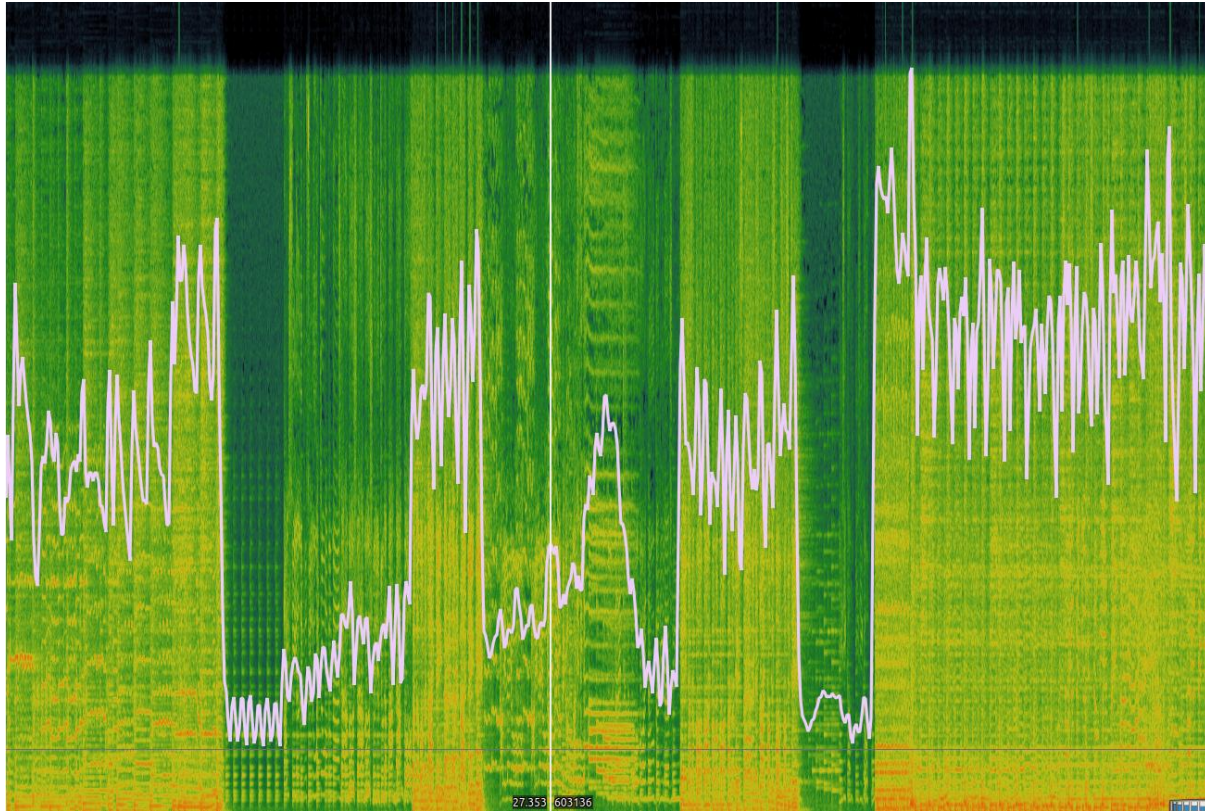
## Low-level spectral features

- Computed directly from the spectrum of the sound
- Very often related to timbre
- Have very little musical meaning and usually non-translatable to higher level musical concepts such as rhythm or harmony



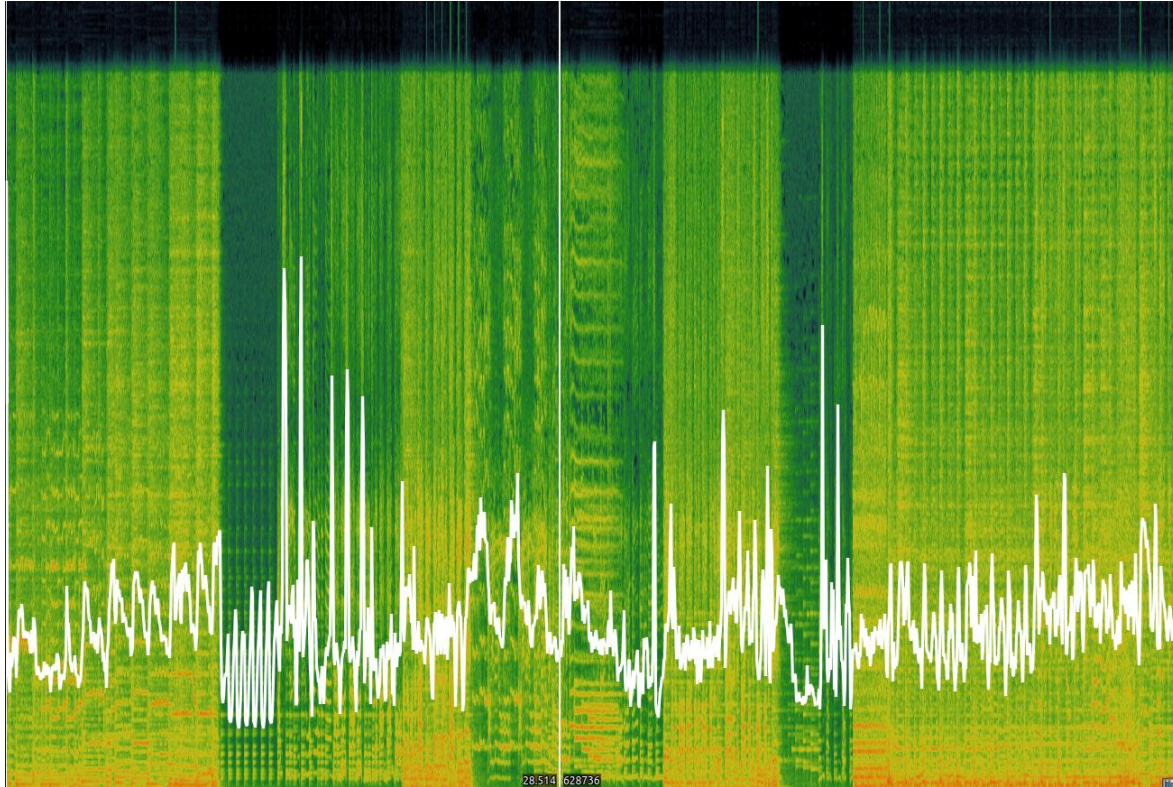


# Spectral energy (RMS)



$$rms = \sqrt{\frac{1}{N} \sum_{i=1}^N x(i)^2}$$

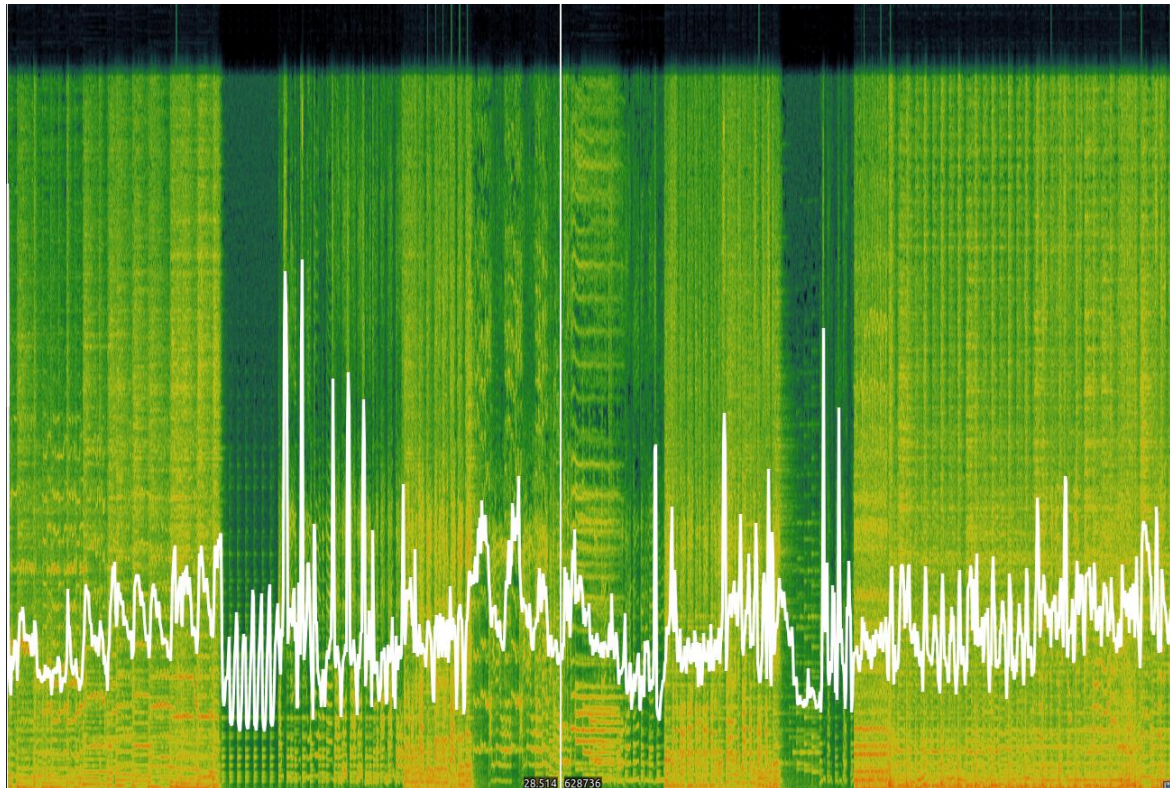
# Spectral centroid



$$\text{Spectral centroid} = \frac{\sum_{n=0}^{N-1} f(n)x(n)}{\sum_{n=0}^{n-1} x(n)}$$

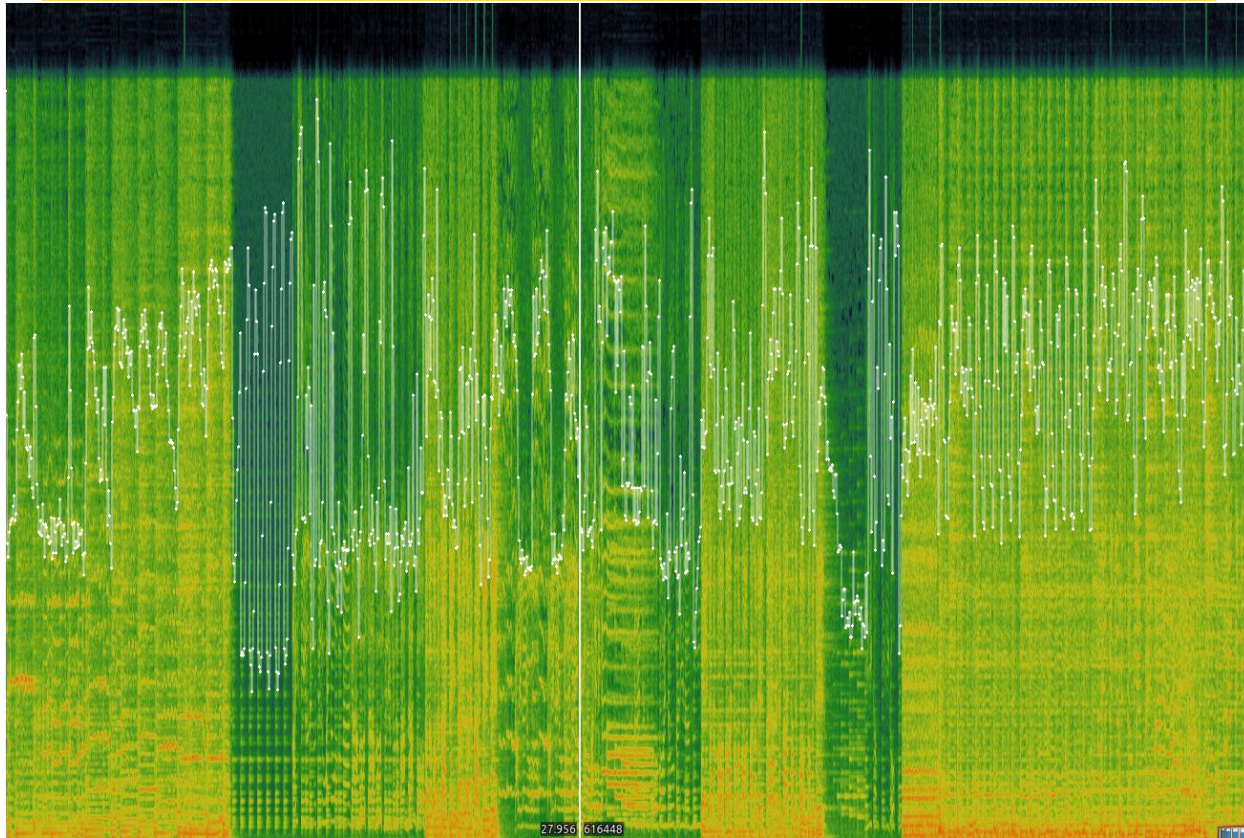


# Spectral kurtosis

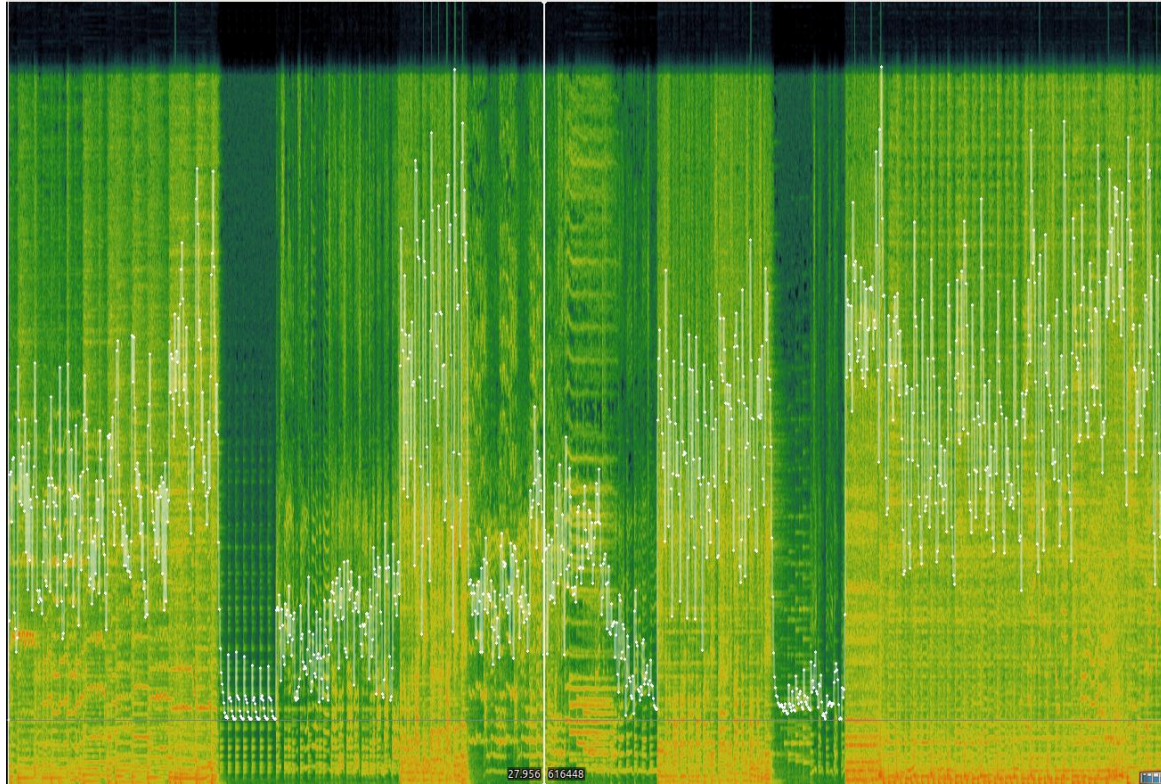




# Spectral roll-off

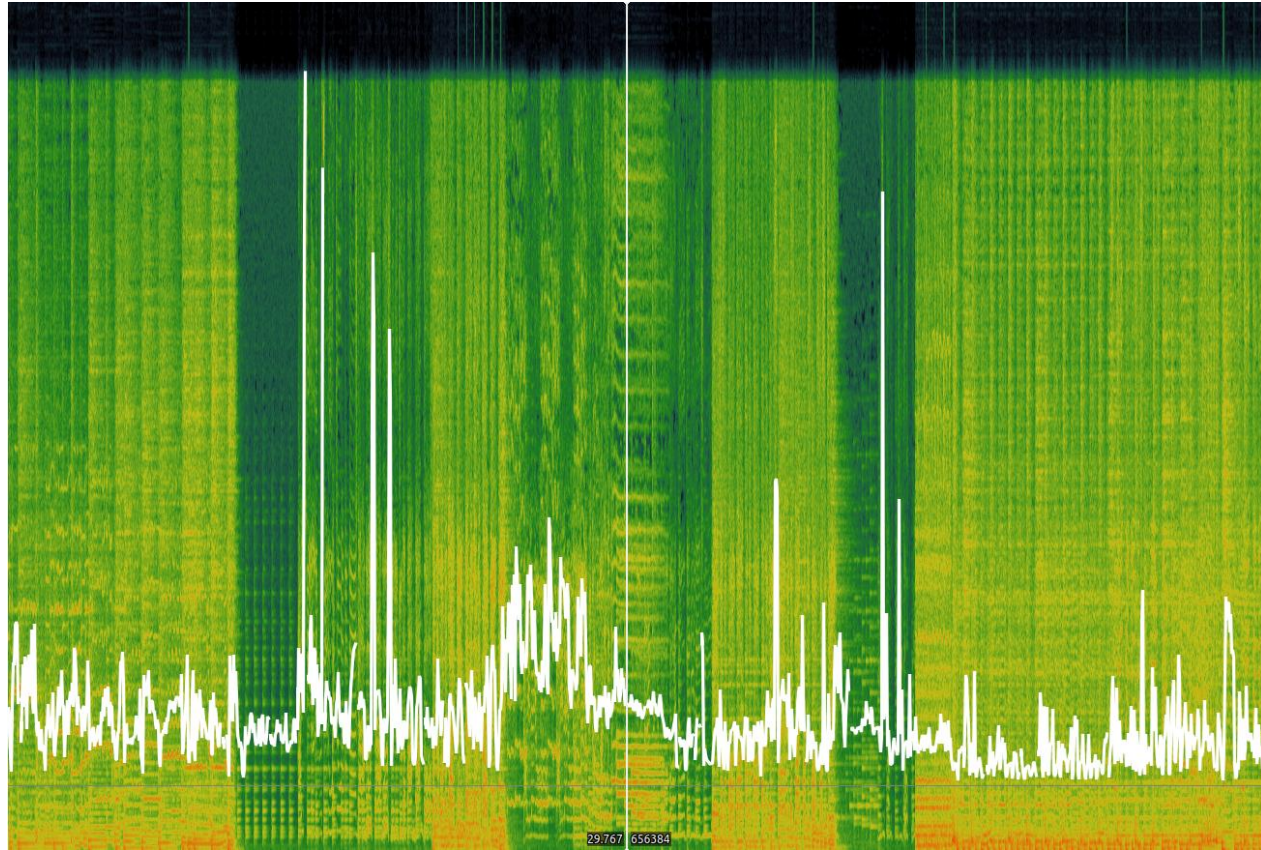


# Spectral flux (between frames)





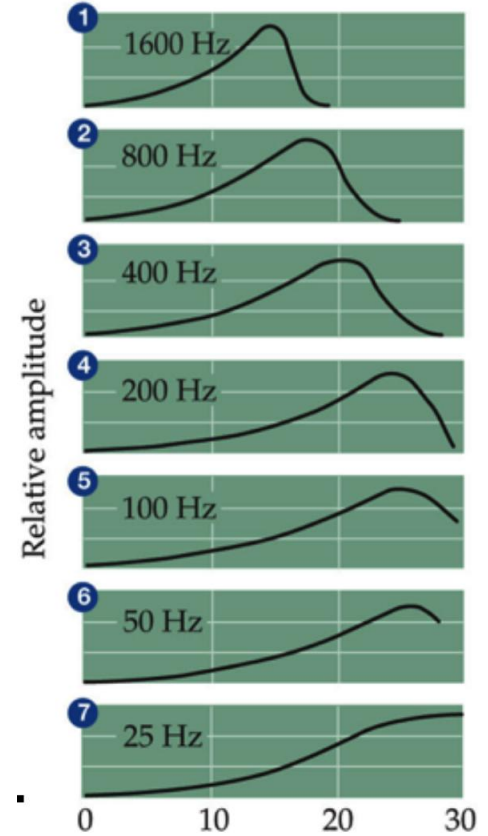
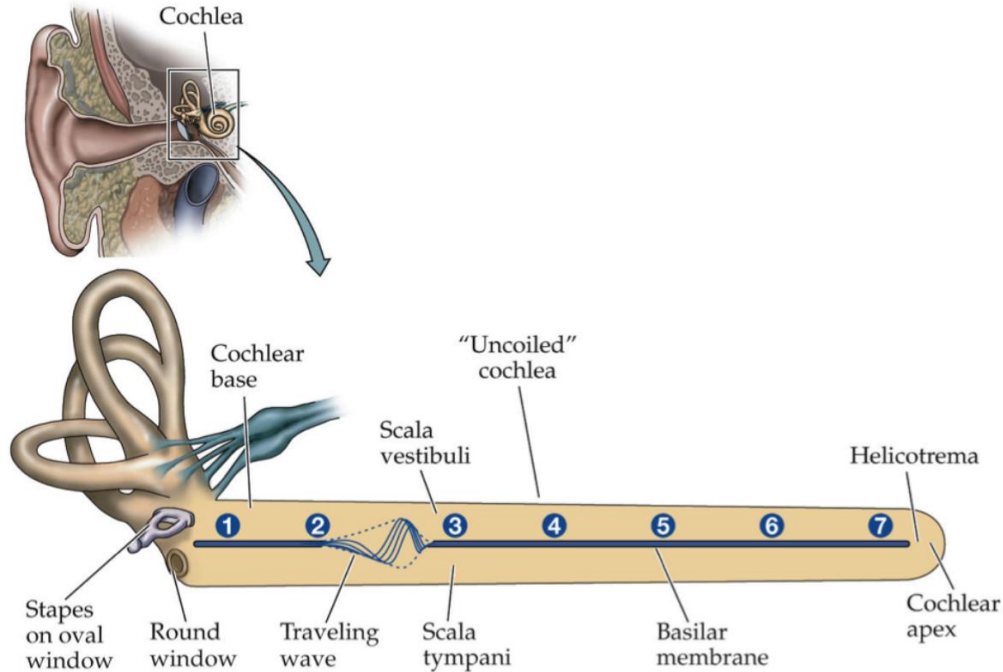
# Inharmonicity



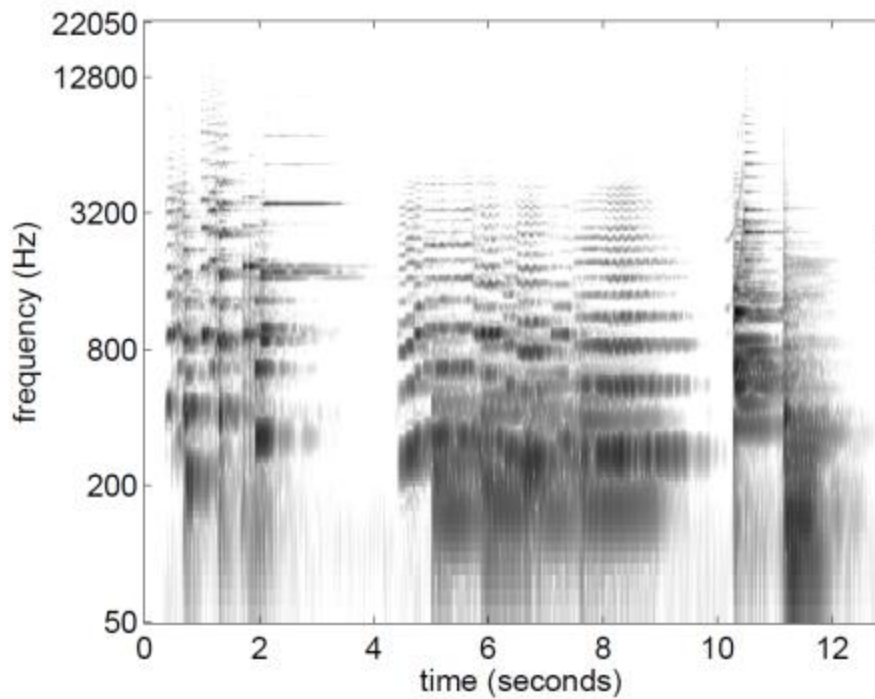
## Perceptual qualities

- We mostly care about the sounds as humans are able to hear them when talking about music
- Logarithmic loudness perception + dependent on the frequency band
- Logarithmic pitch perception (and within boundaries)
- Very attuned to perceiving close pitches or close harmonics of pitches as roughness in sound

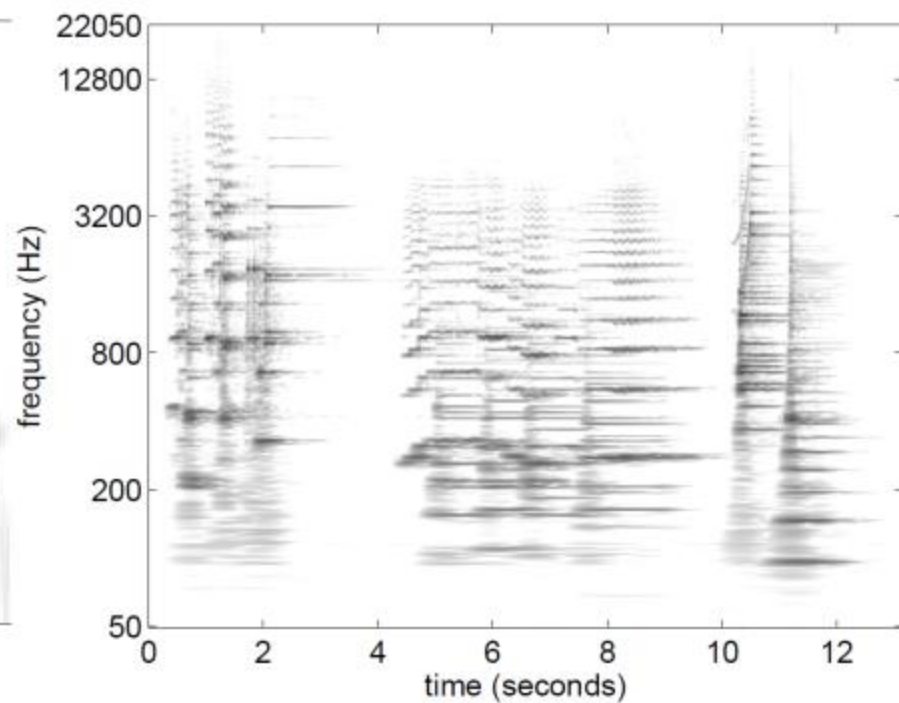
# Loudness perception



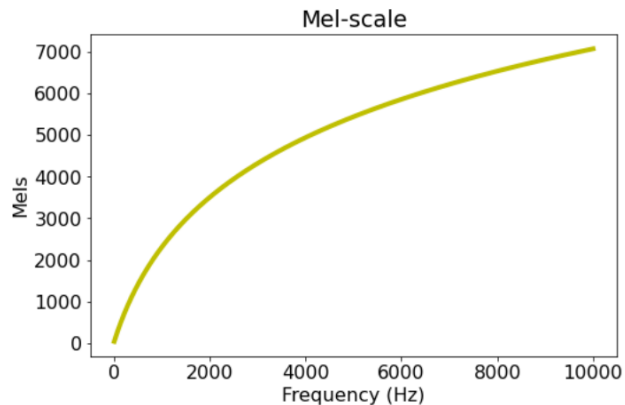
Normal log-scaled STFT spectrogram



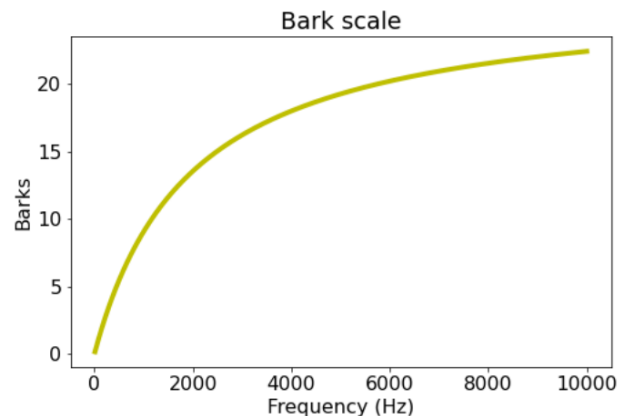
Constant Q spectrogram



## Critical bands of hearing



$$m = 2595 \log_{10} \left( 1 + \frac{f}{700} \right)$$



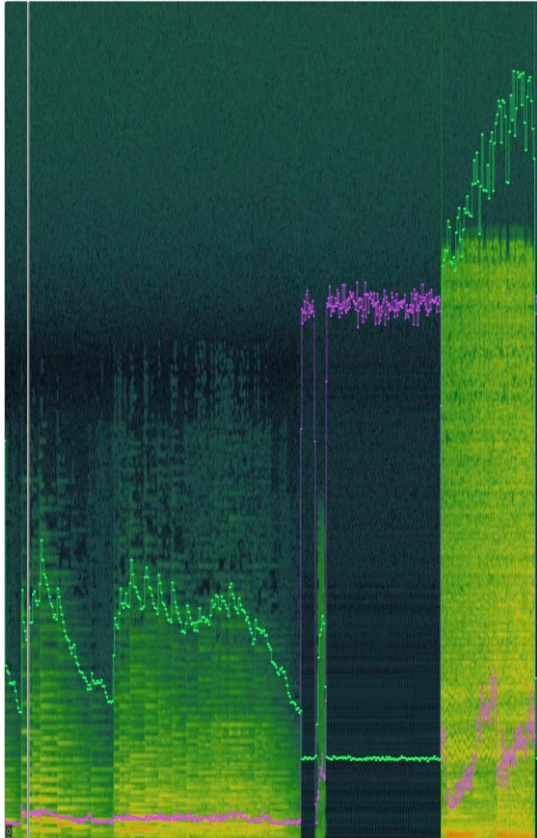
$$\text{Bark} = [(26.81f)/(1960 + f)] - 0.53$$

## Features based on bark scale

- Bark band energy
- Psychoacoustic loudness, normalized by bark bands
- Spectral sharpness (perceptual spectral centroid, related to high frequency content)



## Spectral sharpness



- Spectral sharpness
- Spectral centroid

$$A = 0.11 \cdot \frac{\sum_{z=1}^{nband} z \cdot g(z) \cdot N'(z)}{N}$$

where  $z$  is the index of the band and  $g(z)$  is a function defined by:

$$g(z) = 1 \quad \text{if} \quad z < 15$$

$$g(z) = 0.066 \cdot \exp(0.171z) \quad \text{if} \quad z \geq 15$$

## Spectral roughness

- Perception of temporal envelope modulations in the range of about 20-150 Hz
- Maximal for modulations near 70 Hz
- Perceptual roughness of a complex sound (comprising many partials or pure tone components) depends on the distance between the partials measured in critical bandwidths.
- A simultaneous pair of partials of about the same amplitude that is less than a critical bandwidth apart produces roughness associated with the inability of the basilar membrane to separate them clearly

## Loudness wars

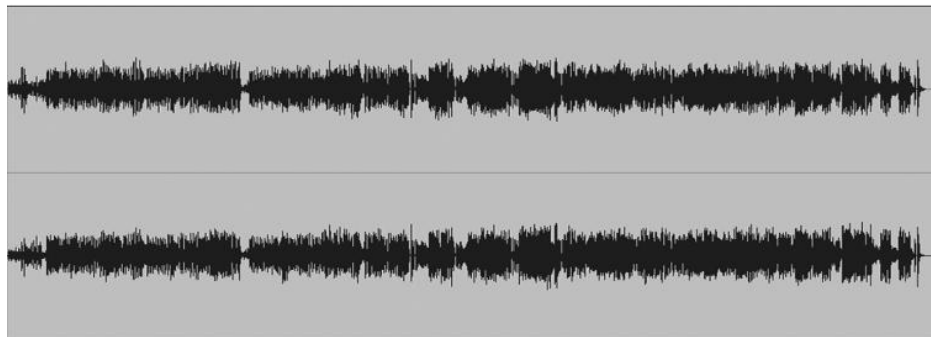
- Humans tend to prefer music that sounds louder
- Unnatural loudness can be achieved by dynamic range compression
- Loudness range audio feature

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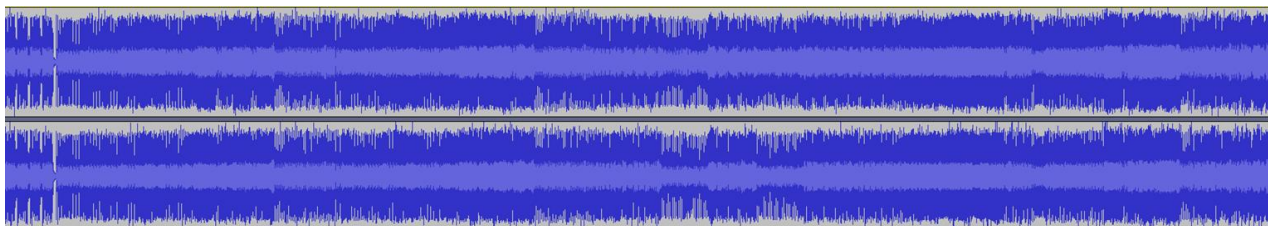
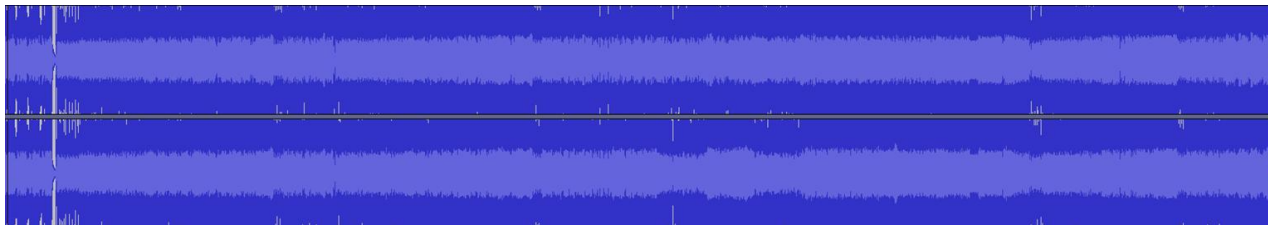
Queen: “Seaside Rendez-vous” (1975)



Red Hot Chili Peppers: “Californication” (1999)



## Metallica remastered album (Death Magnetic)



## Semantic interpretation

Emotion, tagging, similarity, recommendation

## Musical structure

Source separation, segmentation, transcription, instrument detection

## Musical primitives

Onsets, beats, chords

**Model-  
dependent  
abstractions**

**Objective  
measurements  
(little loss)**

## Spectral features

MFCC, ZCR, Bark bands, flux

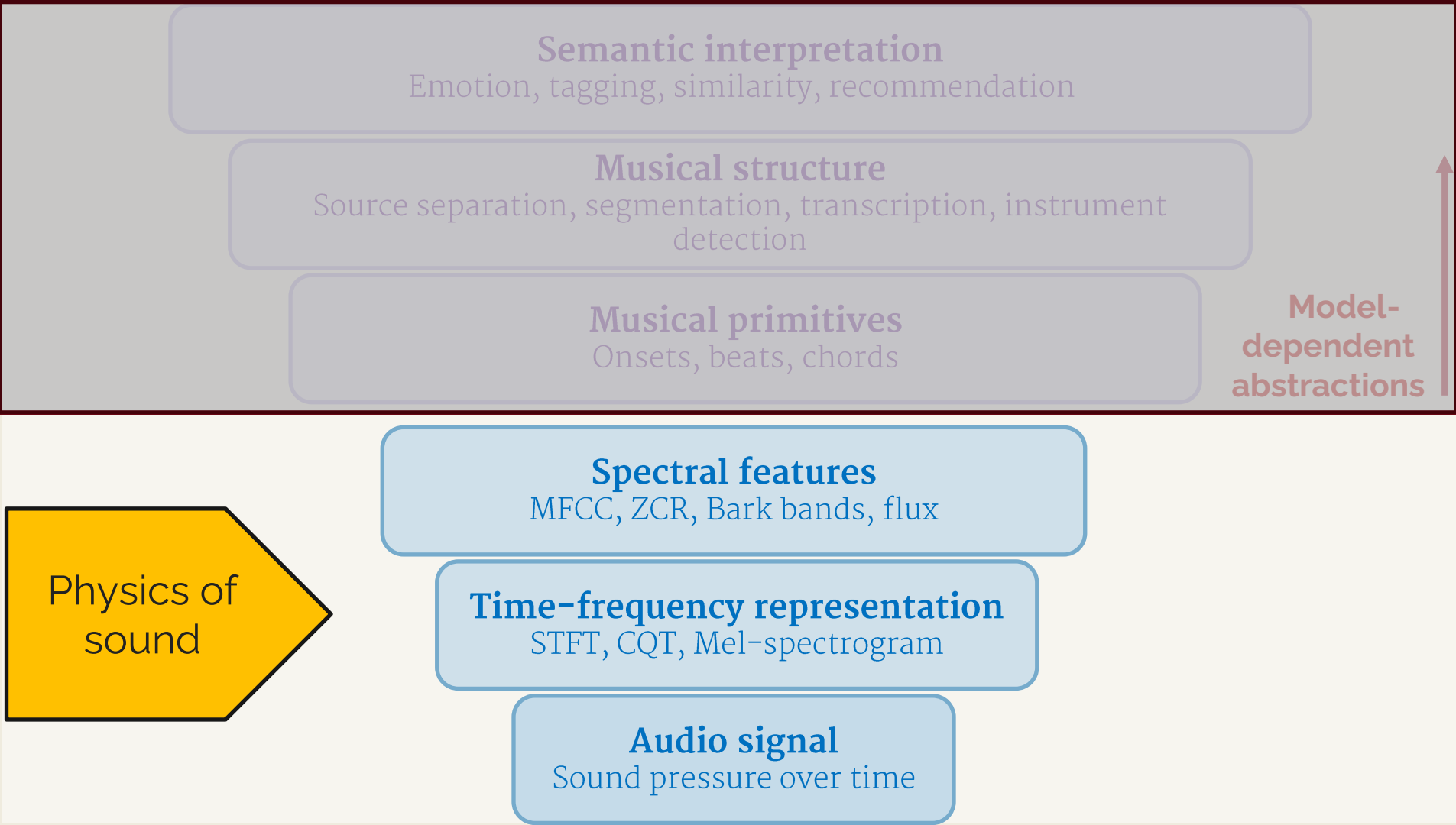
## Time-frequency representation

STFT, CQT, Mel-spectrogram

## Audio signal

Sound pressure over time





Music  
Information  
Retrieval

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Emotion, tagging, similarity, recommendation

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# Music Information Retrieval

